0.1 Exact banded addresses and modeled work

Source item 4.1; module BandedCount

For p packets and half-bandwidth w, the actual address set is

$$\mathcal{B}_{p,w} = \{(i,j) \in \{0,\dots,p-1\}^2 : i \le j+w \text{ and } j \le i+w\}.$$

The predicate is the integer version of $|i-j| \le w$. The count is first defined as a double sum of indicators, then identified with the cardinality of this subtype. It is not defined by the target closed form.

Under $0 \le w < p$, the formal proof establishes

$$\boxed{N_{p,w} = |\mathcal{B}_{p,w}| = p(2w+1) - w(w+1).}$$

The proof uses finite prefix counting, the newly exposed boundary row and column when p increases, and the saturated initial band. An intermediate subtraction-free identity avoids relying on natural-number subtraction without the required inequality. The diagonal case $w = 0$ gives exactly p blocks, and the saturated case $w = p-1$ gives p^2.

Stored scalars. With equal block width b, the storage-address type is

$$\mathcal{S} = \mathcal{B}_{p,w} \times \{0,\dots,b-1\}^2.$$

Its cardinality is the actual number of represented scalar entries:

$$\boxed{|\mathcal{S}| = b^2 N_{p,w}.}$$

Arithmetic events. For q right-hand sides, the declared event type adds a right-hand-side index and one of two operation roles, multiplication or accumulation:

$$\mathcal{O} = \{0,1\} \times \{0,\dots,q-1\} \times \mathcal{S}.$$

Hence

$$\boxed{|\mathcal{O}| = 2qb^2 N_{p,w}.}$$

These identities include zero right-hand-side count. The abstract storage formula also admits zero block width. To use the manuscript's substitution $p = n/b$, one must separately require $b > 0$ and $n = pb$.

Boundary. Two events per coefficient and right-hand side are a declared arithmetic convention. These counts do not identify the number of CUDA instructions, account for fused multiply-add lowering, price complex arithmetic in real operations, or include allocation, indexing, communication and launch overhead. They are also not a performance comparison with dense multiplication.